SEAN PREVITY

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EDUCATION

University of Central Florida, Computer Science, M.S. **2026 — Present**

- GPA: 4.00

University of Central Florida, Computer Science, B.S. **2021 — 2025**

- GPA: 3.64

EXPERIENCE

Software Engineer Intern, YouROK **February 2026 — May 2026**

- Migrated a legacy **PHP** backend to a modular **Node.js/TypeScript** service, redesigning the database schema and implementing a new **GraphQL API**.
- Containerized services with **Docker** and built **CI/CD** pipelines deploying to **DigitalOcean**.
- Eliminated **N+1 query bottlenecks** with **DataLoader** batching and caching, reducing feed requests to a fixed number of database queries regardless of post count.

PROJECTS

Movie Recs **September 2026 — Present**

- Built a film discovery platform with **Next.js**, **Fastify/TypeScript**, and **PostgreSQL** covering **10,000+** films, with filters and similar-film recommendations.
- Ranked similar films via **384**-dim **pgvector** embeddings, weighing theme, genre, and shared cast.
- Added fuzzy search with **pg_trgm** and cut browse query time **95%** with GIN and partial indexes.
- Wrote **64 Vitest** integration tests in **GitHub Actions** CI against a read-only database role.

European Football Stats **October 2025 — Present**

- Built an analytics site for Europe's top **5** leagues with match previews, stats, and lineups.
- Modeled **200+** teams, **3,000+** players, and **35,000+** matches over **20+** years in a relational schema.
- Automated a daily ETL pipeline with **GitHub Actions** that pulls match, odds, and goal-timeline data from **3** external APIs into Postgres.
- Batched H2H lookups for upcoming fixtures, cutting a request from **158** database queries to **2**.
- Moved to a **pnpm** monorepo with **Zod** schemas shared by the **Express** API and **Next.js** client.

France Travel Guide **April — November 2025**

- Designed and built a **bilingual** (French/English) travel site with **Next.js**, **Express/TypeScript**, and **PostgreSQL** with **Drizzle ORM** containing **37,000+** rows of public INSEE data.
- Reduced latency by **80%** and cut API costs for cached cities by implementing an AI description cache.
- Deployed on **AWS** with **API Gateway**, **RDS**, and **EC2**, using **Amplify** and **Cognito** for hosting and authentication.

Babble **January — July 2025**

- Developed a **React/FastAPI** translation platform with layout-preserving DOCX/PDF parsing.
- Benchmarked and **fine-tuned** open-source translation models with **HuggingFace**, improving BLEU scores by up to **35%**.

SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, SQL
- **Frameworks & API:** React, Next.js, Node.js, Express.js, Fastify, FastAPI, GraphQL, Redux, Zod
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase, pgvector, Drizzle ORM
- **Cloud:** AWS (EC2, VPC, API Gateway, Cloud Map, RDS, Amplify, Cognito), DigitalOcean
- **DevOps & Tools:** Docker, GitHub Actions, Linux, Git, Vitest, Postman, Jira